image source: inc.com

BA305 Business Decision-Making with Data

Prof. Mohannad Elhamod

Prof. Gerry Tsoukalas

Boston University Questrom School of Business

General Guidelines

BOSTON UNIVERSITY

Boston University Questrom School of Business

Who is your professor?

NASA

Microsoft Azure

COLLEGE OF ENGINEERING
COMPUTER SCIENCE
VIRGINIA TECH.

Why are you here?

- I am sure it eventually leads to money….
- Maybe it is just a required class?

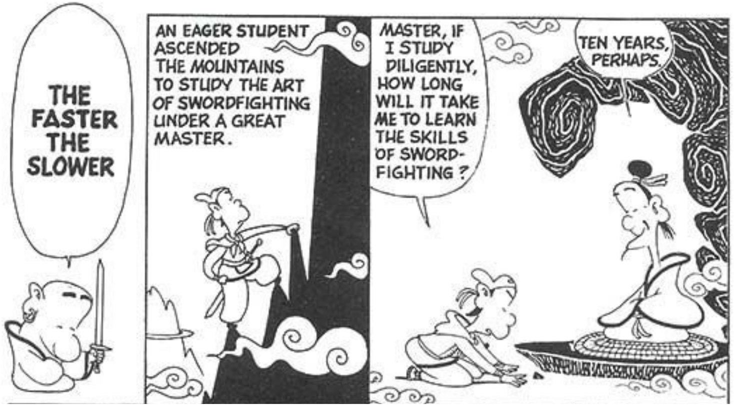
- Nonetheless, understanding how something works is essential for gauging its potential and limitation.

Euclid

> Someone who had begun to [study] geometry asked Euclid, 'What shall I get by learning these things?' Euclid called his slave and said, 'Give him [some money], since he must make gain out of what he learns'.
>
> (Heath, 1981, loc. 8625)

Fundamentals are important!

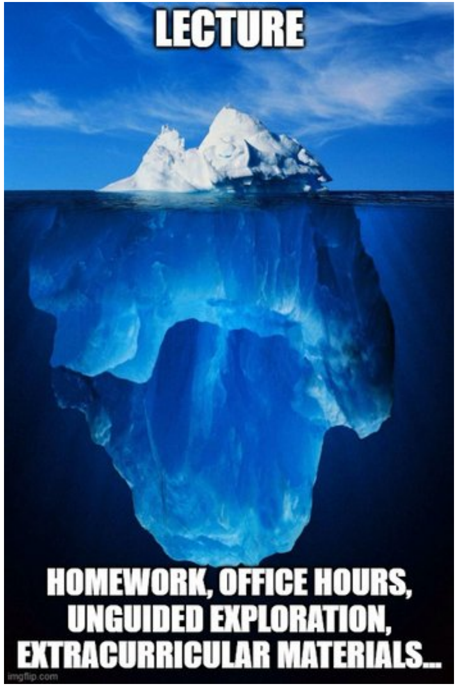
Zen Speaks: Shouts of Nothingness



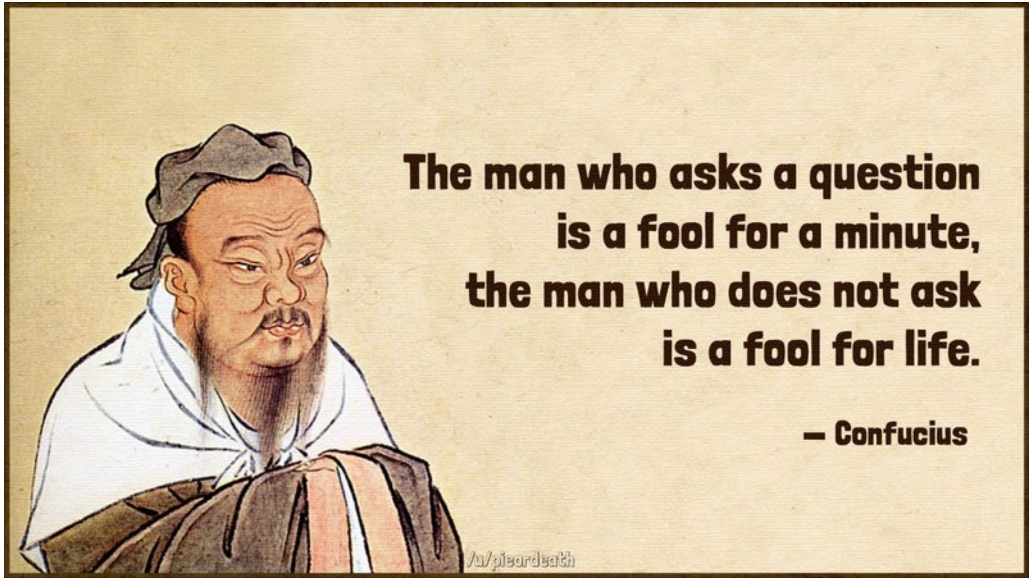
- Skip now, pay later!
- Foundations are especially important.
- Work hard. <u>**Be patient!**</u>

Knowledge Neither Begins Nor Ends in the Classroom

- We build on top of what you have already learned in this program and even high school.
- You are expected to put in the work outside the classroom
  - Attend to your pre-class work and readings
  - Come to office hours.
  - Experiment with code and parameters for fun!
- You are graduate students. Expected to take responsibility for learning!

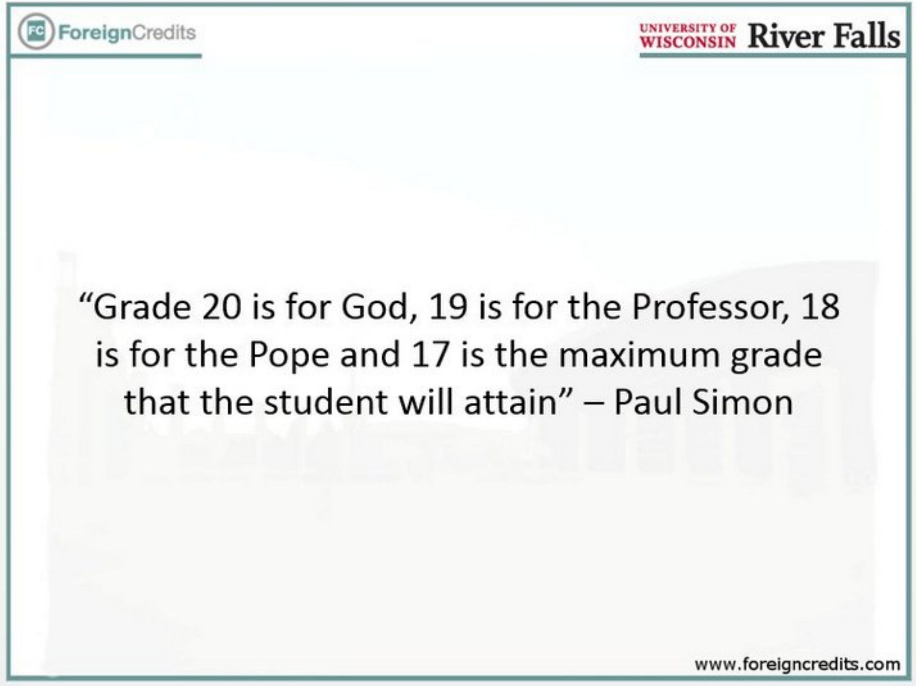


No Question is Foolish



Can I get an A?

- Questrom policy
- But you will get your fair chance.



Participation is Essential!

- 20% of your grade.
- Expectation:
  - Explaining, defending, or countering a position *with sound reasoning*.
  - Discussing and connecting ideas.
  - Looking at a problem from different lenses.
  - It is *NOT* about being correct.

Office Hours

- Meaning: “I did not understand” is not an excuse!
- A project-related question? Sync with your teammates in advance.
- Individual-work question? Do your homework and come with specific questions and proposed pathways.
  - A simple “How do I solve this?” is not a valid question.

Pass through the Syllabus

- Have you actually read it?!
- Pay attention to Piazza for announcements!
- No eating or drinking. Keep laptop and phone screens down unless instructed otherwise.
- ***<u>Zero Tolerance for Academic Integrity Violations!</u>***

Motivatio
n

What is this all about?

Data

Text
Spreadsheet
Time series
…

How?

ML and
Data Analytics
Methods

Linear Regression, Decision Trees,
Neural Nets, Linear Programming…

Decision

Buy/sell
Fraud detection
Optimal pricing
…

The Different Levels of Analytics

- Descriptive: *What happened?...* We can find out through statistics and visuals
- Predictive: *What will happen?...* ML methods. The bulk of this course!
- Prescriptive: *How to make it happen?* Optimization methods.

Example: Analytics at Target

TARGET hired an analyst with the following mandate:

Can you help us identify pregnant women?

How Target Used Data Analytics to Predict Pregnancies

by George Kuhn

Posted at: 1/8/2023 1:30 PM

https://www.driveresearch.com/market-research-company-blog/how-target-used-data-analytics-to-predict-pregnancies/

Example: Analytics at Target

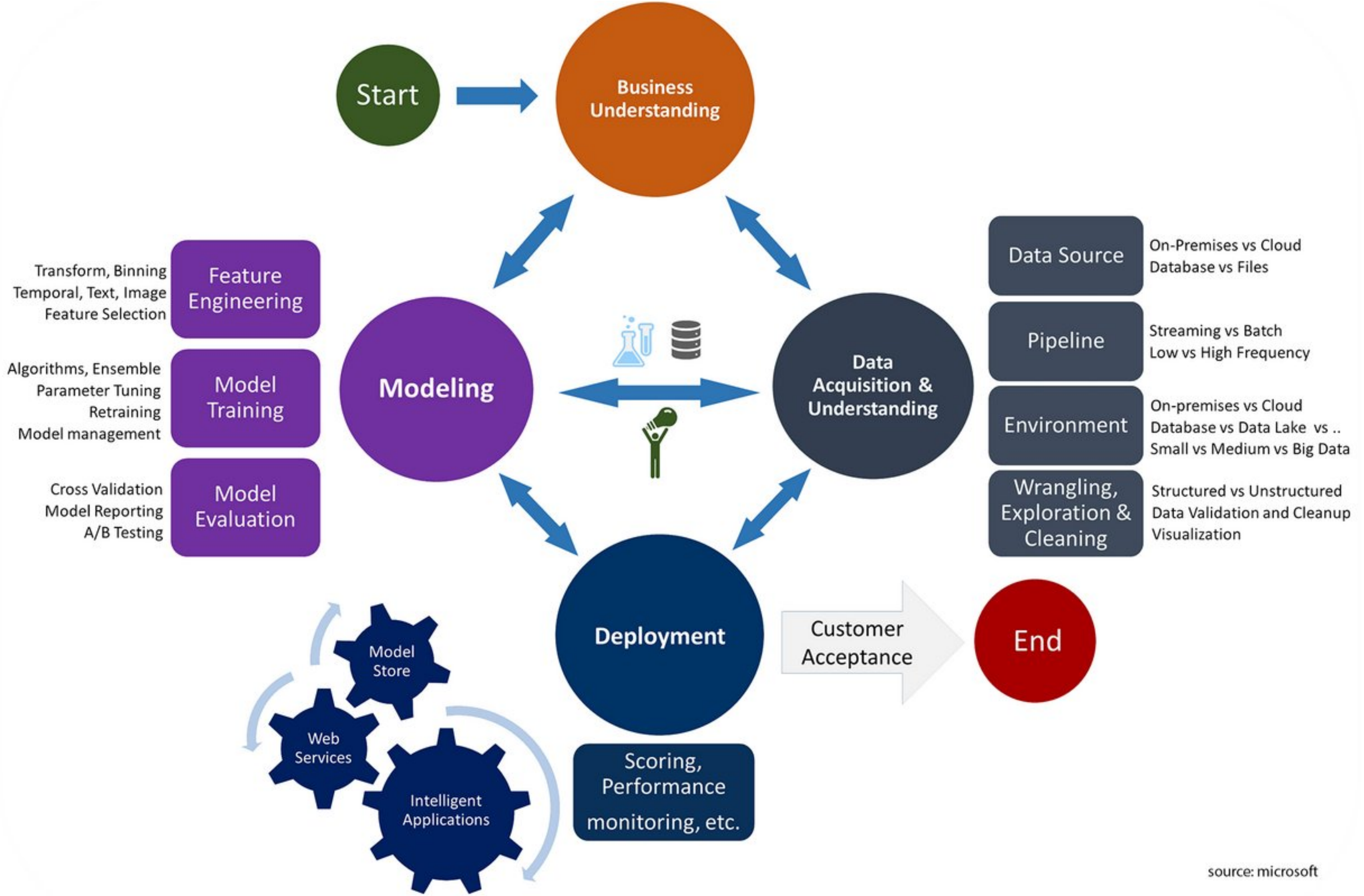
What happened next?

- Public backlash caused Target to “randomize” recommendations
  - Alongside diapers, recommend wine!

- Major impact on Target’s bottom line results
  - CEO commented on the success of the program in their annual report that impressive revenue growth was partially attributable to

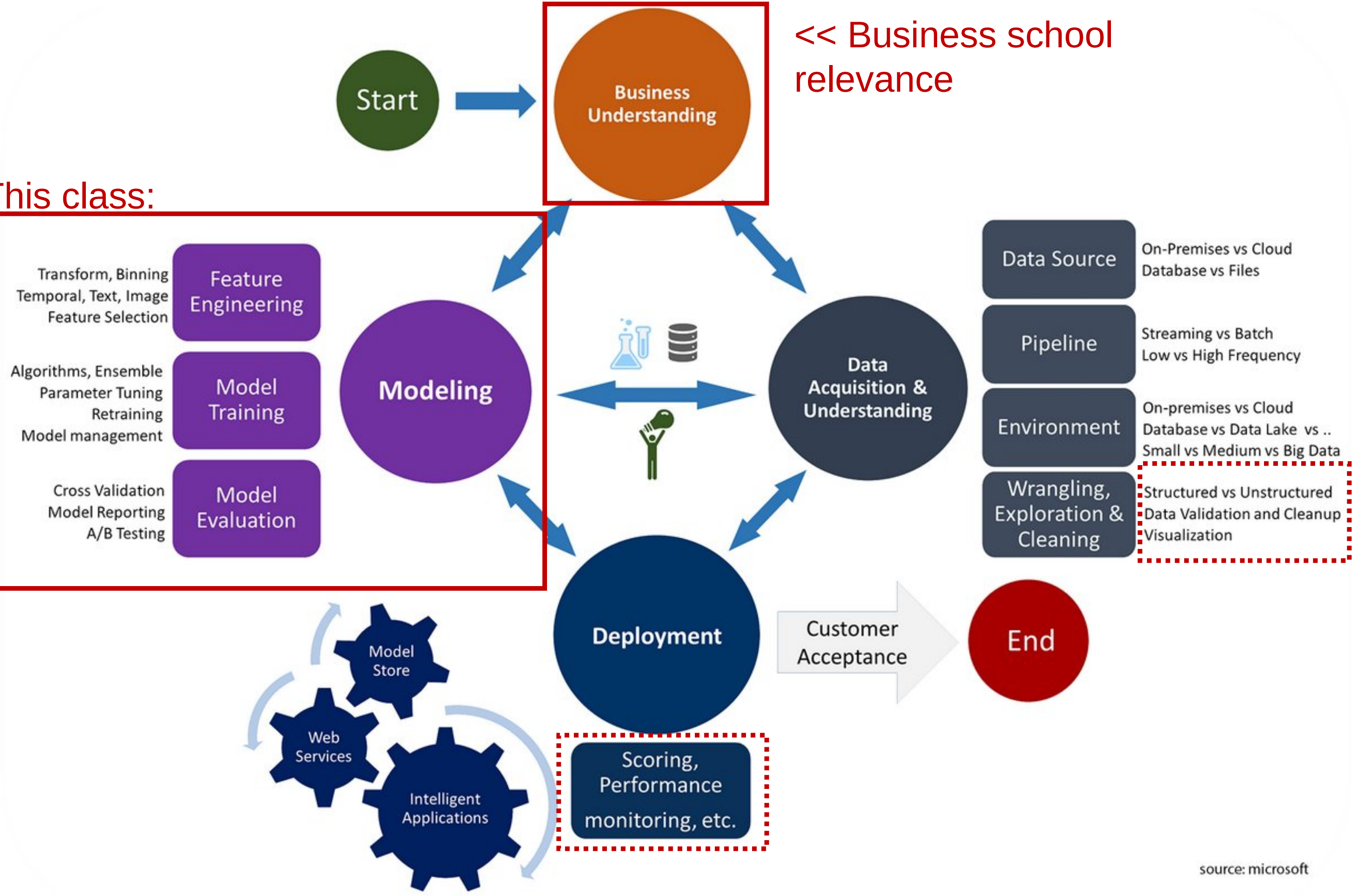
    “heightened focus on items and categories that appeal to specific guest segments such as mom and baby.”

Typical BA Flowchart


Start
Business Understanding
Data Acquisition & Understanding
Data Source
On-Premises vs Cloud
Database vs Files
Pipeline
Streaming vs Batch
Low vs High Frequency
Environment
On-premises vs Cloud
Database vs Data Lake vs ..
Small vs Medium vs Big Data
Wrangling, Exploration & Cleaning
Structured vs Unstructured
Data Validation and Cleanup
Visualization
Modeling
Feature Engineering
Transform, Binning
Temporal, Text, Image
Feature Selection
Model Training
Algorithms, Ensemble
Parameter Tuning
Retraining
Model management
Model Evaluation
Cross Validation
Model Reporting
A/B Testing
Deployment
Model Store
Web Services
Intelligent Applications
Scoring, Performance monitoring, etc.
Customer Acceptance
End
source: microsoft

<< Business school relevance

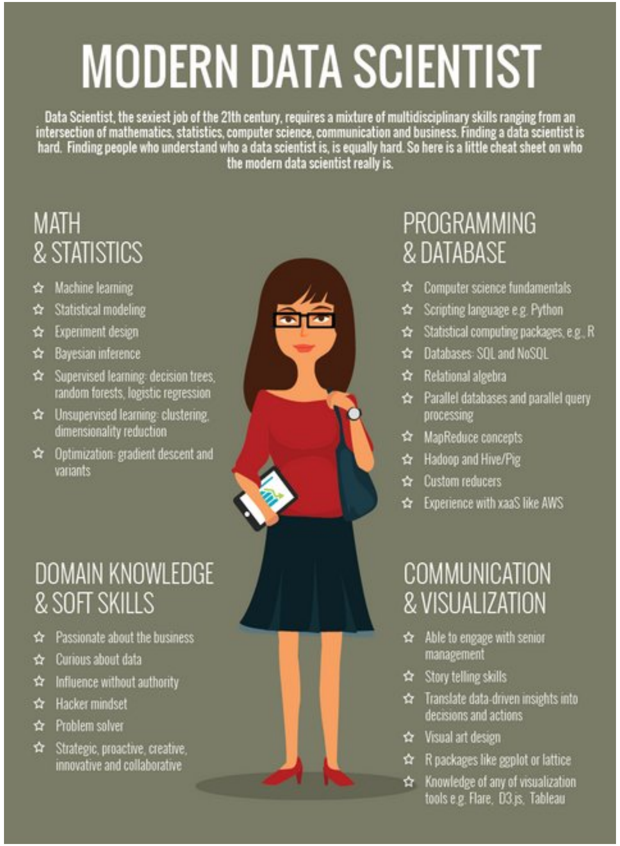
This class:



source: microsoft

The Data Science "Job"

- Very few people master all of these skills simultaneously
- Tip: for your group projects:
  try to mix and match!



Should you still learn to code?

- Zuckerberg Sep 2024
  29:32 – 30:23
  https://www.youtube.com/watch?v=oX7OduG1YmI&t=11s

- *“One of the big debates is like should we still teach our kids to program computers even though you're going to have these tools in the future that are just so much more powerful than anything than we have now to produce incredible complicated pieces of software.”*

- *“I think the answer to that is probably yes because I think teaching someone how to code is teaching them a way to think rigorously. And even if they are not doing most of the code production, I think it's important that you have the ability to think in that way, and it's going to make you a better thinker and better person. Maybe that's this generation's version of calculators.”*

Example: Job posting at Disney+

United States | Jobs

CAREER AREAS | BRANDS | PROGRAMS | ABOUT US

Data Analyst

Job ID: 854719BR
Location: New York, New York, United States
Business: Disney Streaming

Job Summary:

The Disney+ Product & Tech Analytics team is seeking a Data Analyst who will be an outstanding addition to our team. We partner with other teams across the business to turn data into information, information into insights and insights into key business decisions that steer the strategic direction of the company. The right person for this role thrives in a fast-paced environment, loves problem-solving, and is a good communicator; telling high-impact stories with data. If you are someone who gains personal happiness from transforming data into knowledge and driving business action, then this is a phenomenal role for you.
You will design innovative analytic approaches to test hypotheses, extract actionable insights by identifying patterns in structured and unstructured data, and develop recommendations to drive product performance across Disney's streaming platforms. You will be instrumental in leading the use of analytics as a critical value driver across our organization.

United States | Jobs

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Responsibilities:

- Business Performance Reporting, Visualization: Be a partner for cross-functional stakeholders to understand performance of the platform. When a metric is ahead or behind expectations, dig in to drive meaningful, actionable insights that help the business performance improve or double-down on success.
- Deep-Dive Analysis: Use advanced techniques (statistical modeling, experimentation, causal inference) to analyze user behavioral data to identify patterns and uncover opportunities. Create a rich understanding of how people are interacting with the platform and content and use these insights to then make recommendations.
- Opportunity Identification and Sizing: Both to set goals and help evaluate potential opportunities, this analyst will be tasked with forecasting and opportunity sizing. This will help business stakeholders evaluate trade-offs in different approaches, and create targets for driving business performance.
- Partnership and communication: Partner closely with business stakeholders to identify and unlock opportunities, and with other data teams to improve platform capabilities around data modeling, testing platforms, data visualization, and data architecture.

United States | Jobs

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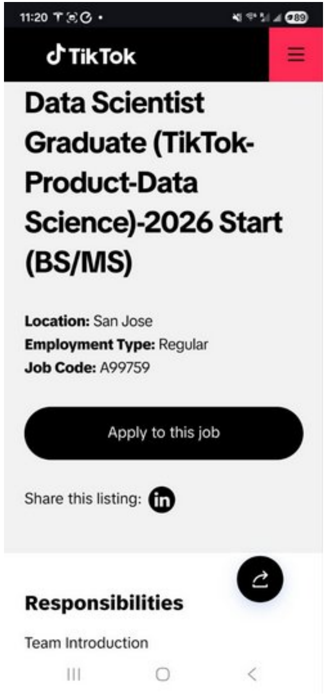
Basic Qualifications:

- A bachelor's or master's degree in Computer Science, MIS, Statistics, Economics, Mathematics, etc.
- 2+ years of analytical experience.
- 2+ years' work experience using SQL
- Some experience with Python/R or other statistical programming language
- Familiarity with data exploration and data visualization tools like Tableau, Looker, Chartio, etc.
- Understanding of statistics concepts (e.g. confidence intervals, hypothesis testing, regression analysis).
- Ability to think strategically, analyze and interpret market and consumer information.
- Intense curiosity, high intellectual and analytical horsepower, passionate pursuit of analytics driven insights
- Strong communication skills, as well as written and verbal presentation skills and experience presenting to diverse audiences.

Preferred Qualifications:

- General knowledge of optimization algorithms (e.g. MAB, randomized search, Thompson sampling), supervised and unsupervised machine learning techniques.

Example: Job posting at TikTok



Data Scientist Graduate (TikTok-Product-Data Science)-2026 Start (BS/MS)

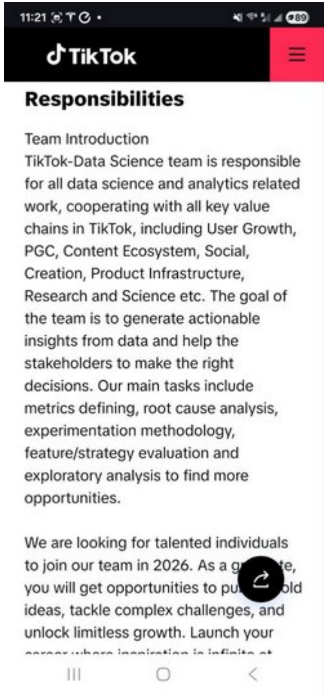
Location: San Jose
Employment Type: Regular
Job Code: A99759

Apply to this job

Share this listing:

Responsibilities

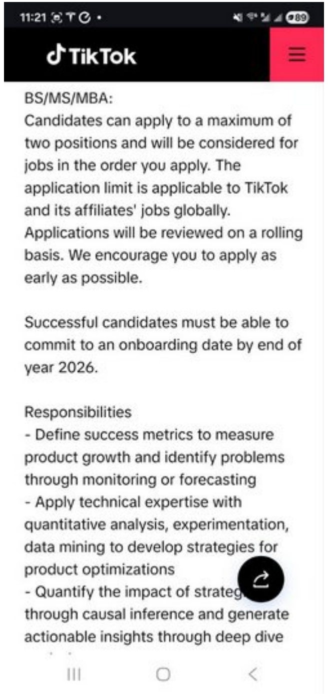
Team Introduction



Responsibilities

Team Introduction
TikTok-Data Science team is responsible for all data science and analytics related work, cooperating with all key value chains in TikTok, including User Growth, PGC, Content Ecosystem, Social, Creation, Product Infrastructure, Research and Science etc. The goal of the team is to generate actionable insights from data and help the stakeholders to make the right decisions. Our main tasks include metrics defining, root cause analysis, experimentation methodology, feature/strategy evaluation and exploratory analysis to find more opportunities.

We are looking for talented individuals to join our team in 2026. As a g[illegible]te, you will get opportunities to pu[illegible]old ideas, tackle complex challenges, and unlock limitless growth. Launch your



BS/MS/MBA:
Candidates can apply to a maximum of two positions and will be considered for jobs in the order you apply. The application limit is applicable to TikTok and its affiliates' jobs globally. Applications will be reviewed on a rolling basis. We encourage you to apply as early as possible.

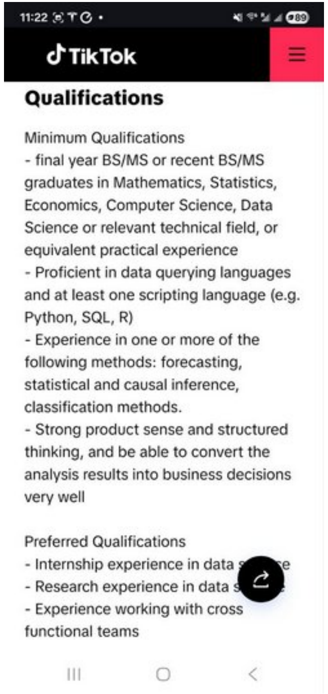
Successful candidates must be able to commit to an onboarding date by end of year 2026.

Responsibilities
- Define success metrics to measure product growth and identify problems through monitoring or forecasting
- Apply technical expertise with quantitative analysis, experimentation, data mining to develop strategies for product optimizations
- Quantify the impact of strateg[illegible] through causal inference and generate actionable insights through deep dive

Responsibilities
- Define success metrics to measure product growth and identify problems through monitoring or forecasting
- Apply technical expertise with quantitative analysis, experimentation, data mining to develop strategies for product optimizations
- Quantify the impact of strategies through causal inference and generate actionable insights through deep dive analysis
- Partner with Product, Engineering and cross-functional teams to influence and execute product strategy

Qualifications

Minimum Qualifications
- final year BS/MS or recent BS/MS graduates in Mathematics, Statistics, Economics, Computer Science, Da[illegible] Science or relevant technical fi[illegible] equivalent practical experience
- Proficient in data querying languages and at least one scripting language (e.g.



Qualifications

Minimum Qualifications
- final year BS/MS or recent BS/MS graduates in Mathematics, Statistics, Economics, Computer Science, Data Science or relevant technical field, or equivalent practical experience
- Proficient in data querying languages and at least one scripting language (e.g. Python, SQL, R)
- Experience in one or more of the following methods: forecasting, statistical and causal inference, classification methods.
- Strong product sense and structured thinking, and be able to convert the analysis results into business decisions very well

Preferred Qualifications
- Internship experience in data s[illegible]e
- Research experience in data s[illegible]
- Experience working with cross functional teams

Data

Data representation

- Data can often be represented or abstracted as an r (rows) × c (columns) matrix
  - Rows correspond to instances in the dataset
  - Columns represent features or properties of interest

supervised

$$
\begin{array}{c|c}
 & \begin{array}{ccccc} X_1 & X_2 & \dots & X_c & Y \end{array} \\
\hline
\begin{array}{c} \mathbf{x_1} \\ \mathbf{x_2} \\ \vdots \\ \mathbf{x_r} \end{array} &
\begin{bmatrix}
x_{11} & x_{12} & \dots & x_{1c} & y_1 \\
x_{21} & x_{22} & \dots & x_{2c} & y_2 \\
\vdots & \vdots & \ddots & \vdots & \vdots \\
x_{r1} & x_{r2} & \dots & x_{rc} & y_r
\end{bmatrix}
\end{array}
$$

unsupervised

$$
\begin{array}{c|c}
 & \begin{array}{cccc} X_1 & X_2 & \dots & X_c \end{array} \\
\hline
\begin{array}{c} \mathbf{x_1} \\ \mathbf{x_2} \\ \vdots \\ \mathbf{x_r} \end{array} &
\begin{bmatrix}
x_{11} & x_{12} & \dots & x_{1c} \\
x_{21} & x_{22} & \dots & x_{2c} \\
\vdots & \vdots & \ddots & \vdots \\
x_{r1} & x_{r2} & \dots & x_{rc}
\end{bmatrix}
\end{array}
$$

Data representation

- Depending on the application domain
  - rows may be referred to as entities, instances, examples, records, transactions, objects, points, feature vectors, tuples…
  - columns may be called attributes, properties, features, dimensions, variables, fields...
- The number of instances, r, is referred to as the size of the data, whereas the number of features, c, is called the dimensionality of the data

c=1 is Univariate analysis
c=2 is Bivariate analysis
c>2 is Multivariate analysis

- Different dimensions imply different tools…

Example: Data matrix

Projection of x Load	Projection of y load	Distance	Load	Thickness
10.23	5.27	15.22	2.7	1.2
12.65	6.25	16.22	2.2	1.1

Example: Transaction Data

- Each row is a customer transaction
  - Not as easy to put in matrix form…

TID	*Items*
1	Bread, Coke, Milk
2	Beer, Bread
3	Beer, Coke, Diaper, Milk
4	Beer, Bread, Diaper, Milk
5	Coke, Diaper, Milk

Example: Sequence Data

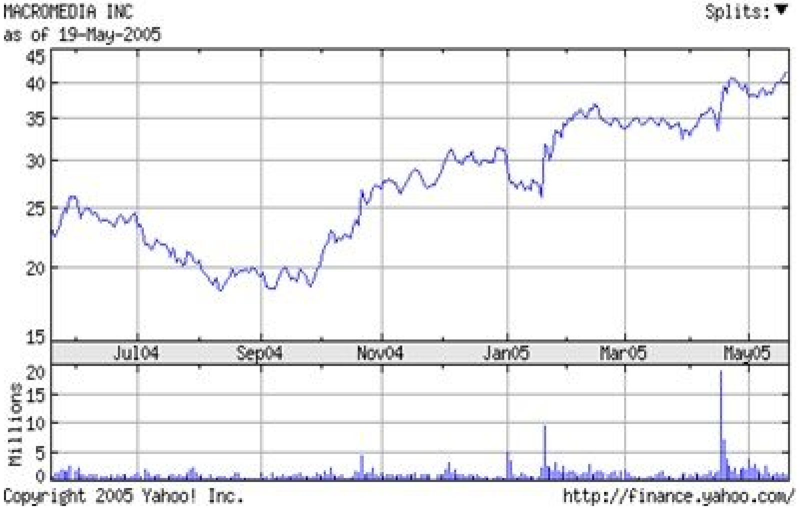
- Genomic sequence data

GGTTCCGCCTTCAGCCCCGCGCCC
GCAGGGCCCGCCCCGCGCCGTCA
GAAGGGCCCGCCTGGCGGGCGGG
GGGAGGCGGGGCCGCCCGAGCCA
ACCGAGTCCGACCAGGTGCCCCCT
CTGCTCGGCCTAGACCTGAGCTCAT
TAGGCGGCAGCGGACAGCCAAGTA
GAACACGCGAAGCGCTGGGCTGCC
TGCTGCGACCAGGG

Example: Time Series

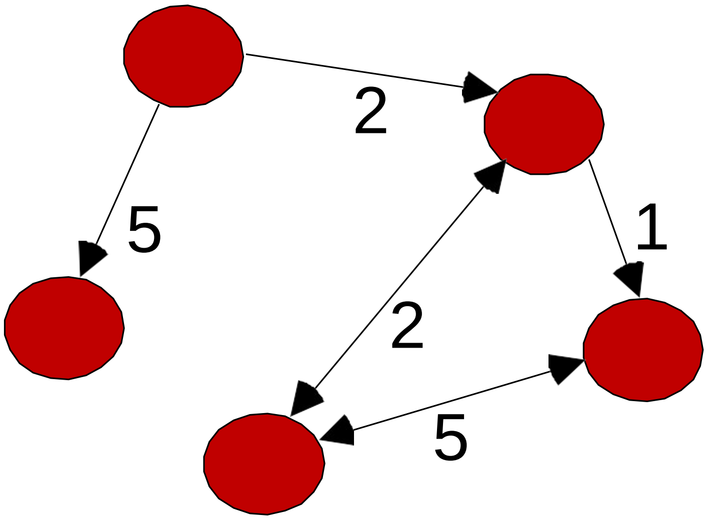
ECG Heartbeat

Stock



Example: Graph Data

- Generic graph and HTML links



Data Mining

Graph Partitioning

Parallel Solution of Sparse Linear System of Equations

<a href="papers/papers.html#ffff">
N-Body Computation and Dense Linear System Solvers

Types of Data and Measurement Scales

Data

- **Metric (Quantitative)**
  - **Ratio Scale (e.g., prices of products)**

    e.g., a weight of 0lbs. means no weight

    Can calculate differences and ratios
  - **Interval Scale (e.g., temperature in Fahrenheit)**

    e.g., a temp. of 0 does not mean no temperature.

    Can calculate differences, not ratios
- **Non-Metric (Qualitative)**
  - **Ordinal Scale (e.g., class ranking)**

    e.g., ranking 1st, is better than 2nd.

    But intervals may not be consistent, so differences are not meaningful
  - **Nominal Scale (e.g., color, blood type)**

    e.g., red isn't "more than" blue

    Can count frequencies but not order them

Example

- Real world data is usually "dirty"
  - Incomplete data
    - (missing values, only aggregated data, etc.)
  - Inconsistent data
    - (different coding, impossible values or out-of-range values)
  - Noisy data
    - (data containing errors, outliers, not accurate)

A mistake or a millionaire?

Missing values

Inconsistent duplicate entries

Tid	Refund	Marital Status	Taxable Income	Cheat
1	Yes	Single	125K	No
2	No	Married	100K	No
3	No	Single	70K	No
4	Yes	Married	120K	No
5	No	Divorced	10000K	Yes
6	No	NULL	60K	No
7	Yes	Divorced	220K	NULL
8	No	Single	85K	Yes
9	No	Married	90K	No
9	No	Single	90K	No

Data Preprocessing: Examination phases

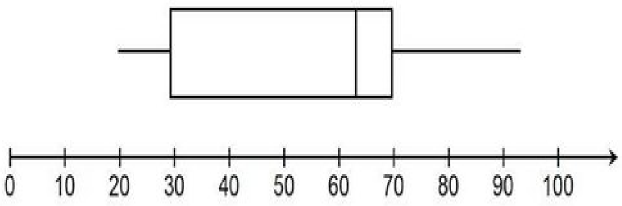
- Data Preparation: It is important to examine the suitability of the data before the analysis and make the necessary transformations and adjustments. The care taken during this step directly affects the result of the modeling process. It is not uncommon for data preparation to take up to **80% of the modeling effort**.

- Visualization/graphical examination
- Identify and evaluate <u>missing values</u>
- Identify and deal with <u>outliers</u>
- Develop a preliminary understanding of your data

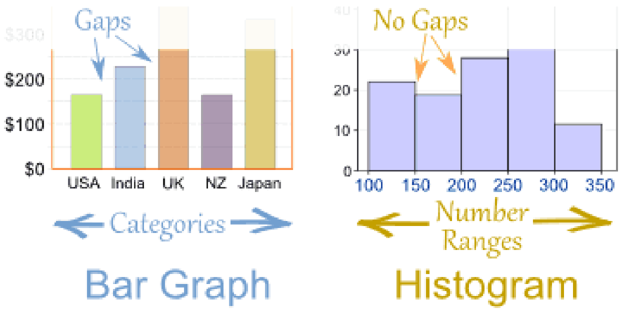
Data Preprocessing: Graphical examination

- Shape
  - Histogram
  - Box & Whisker plot

- Relationships
  - Scatterplot
  - Outliers



Box Plot



Scatter Plot

What to do about missing data

- Why is it missing?
  - There is a systematic reason.
  - For no particular reason…

- What to do?
  - <u>Remove</u> the records with missing data.
  - <u>Impute</u> the missing values (Replace with mean or some default value. There are other more complex methods).

- Impact of missing data treatment:
  - Reduces sample size available for analysis.
  - Can distort results.

Outliers

An observation/response with a unique combination of characteristics identifiable as distinctly different from the other observations/responses. The issue is whether the observation/response is representative of the population.

- Why Do Outliers Occur?
  - An error
  - An extraordinary/unique observation
- Dealing with Outliers
  - Identify outliers:
    - Using standard deviation, which can be done visually (box plots) or numerically.
    - Other more complex methods.
  - Delete or retain?

Data Transformations

- Data transformations can help clean up the data and produce better analytical results.
  - <u>Normalization:</u> (remove mean, divide by standard deviation)
  - <u>Aggregation:</u> Can help reduce data size and get rid of noise. Can be performed both at the record or attribute levels.

Data Transformations: Dummy Variables

- Categorical format (nonmetric):

Client	Profession
1000	Physician/0
1001	Attorney/1
1002	Professor/2

- Dummy Variable format (metric):

Client	Physician	Attorney	Professor
1000	1	0	0
1001	0	1	0
1002	0	0	1